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Regular Article



Leisure, emotional solidarity and behavioural intentions in academic library. the moderating roles of social identity and gender

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ABSTRACT

This study examined the effects of leisure activities and emotional solidarity on students' intention to use a library, by considering the moderating effects of social identity and gender. This study provides insights for libraries to improve their spaces and services to attract and assist all students effectively. A quantitative survey was conducted with 838 students in Ghana to measure their engagement in leisure activities, emotional connectedness in the library, social resources, gender, and inclination to use the library for academic purposes. The results showed that engagement in recreational activities, emotional connectedness, and social resources positively influenced students' intention to use the library. However, the relationship between leisure activities and library utilisation was weaker for students with more social capital, suggesting the need for customised leisure activities that cater to social needs. Males showed greater intentions to use the library than did females; however, this gender disparity decreased for females with strong emotional attachments to the library, highlighting the importance of fostering a sense of belonging for female students. These findings suggest that libraries can increase student utilisation by integrating recreational activities, promoting emotional unity, and addressing social and gender needs. Creating a welcoming and interactive atmosphere that encourages social engagement, a sense of belonging, and assistance can enhance the academic achievement of all students. Further research on causal relationships and specific strategies for implementing these findings can provide valuable insights into library establishment and management.

1. Introduction

Academic libraries have undergone a substantial transformation over the past decade. They are no longer confined to their traditional roles as silent information repositories. Increasingly, these spaces are evolving into dynamic environments that combine academic learning, emotional support, and social engagement (Tachie-Donkor & Ezema, 2023). This transition reflects a broader shift in higher education, where students expect environments that not only support academic tasks, but also foster psychological well-being, leisure, and a sense of belonging. Although there is growing interest in how libraries contribute to student life, limited attention has been paid to how students interact with these multifaceted environments, especially in contexts where academic and social needs converge (see Fig. 1).

A major gap in the literature lies in the limited understanding of how leisure activities and emotional solidarity influence students' behavioural intentions to use academic libraries for academic purposes. Many existing studies treat library use for academic and recreational purposes as separate phenomena (Alsalhi et al., 2020; Liu et al., 2022). This separation overlooks the integrated nature of students' experiences in modern libraries, where leisure and emotional interactions often precede or support academic engagement. Emotional solidarity, defined as feelings of connection, trust, and belonging to students' experiences in academic communities, has rarely been examined as a factor influencing library-related academic behaviour (Woosnam & Aleshinloye, 2013). Likewise, social identity and gender, which may influence how students relate emotionally to the library space and its community, have not been adequately explored in previous studies, particularly in the sub-Saharan African educational context. This oversight limits the understanding of how psychosocial and demographic factors shape student-library interactions.

The absence of such research has practical implications for academic institutions. Without understanding the interplay between leisure engagement, emotional connections, and academic behaviour, libraries

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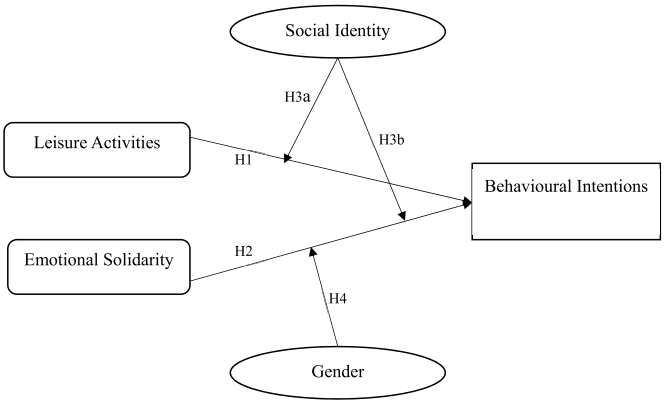


Fig. 1. The study model.

may fail to serve students' holistic needs. Emotional solidarity can affect whether students, especially female students, feel welcomed and sufficiently supported to use library spaces for academic purposes (Zhang et al., 2023). Similarly, students who strongly identify with the library community may be more motivated to participate in academic tasks in that setting. Neglecting these emotional and social variables risks overlooking key barriers to engagement, and may limit the effectiveness of student support systems. This concern is particularly relevant in Ghana, where academic libraries are increasingly viewed as hubs for both social and academic development.

Over the last ten years, academic library research has focused primarily on digital transformation, user satisfaction, and information-seeking behaviours (Cox & Brewster, 2020; DeVille & Sughrue, 2023). While these areas are important, they rarely address the social and emotional factors that influence students' behavioural intentions in library settings. This study addresses this gap by drawing on three well-established theories: social identity theory, the theory of planned behaviour, and the uses and gratifications theory. Social Identity Theory explains how affiliation with a social group, such as a library community, fosters emotional solidarity and influences behavioural motivation. The Theory of Planned Behaviour offers insight into how perceived norms and behavioural control shape students' academic intentions (Ajzen, 2011). The uses and Gratifications Theory helps explain the role of leisure activities in satisfying students' need for relaxation, social interaction, and academic stimulation (Williams, 2021).

The novelty of this research lies in its integrated examination of leisure activities, emotional solidarity, social identity, and gender in predicting behavioural intentions toward library use. Unlike most previous studies that focused on individual variables or Western contexts, this study explores how these psychosocial and demographic factors interact in the context of Ghanaian higher education. This approach allows for a nuanced understanding of the role of libraries in supporting students' academic development through emotional and social engagement. This study contributes to theoretical development and offers practical insights for library managers and university administrators seeking to design inclusive and engaging learning environments.

Based on the foregoing discussion, this study was guided by the following research question.

1. How do leisure activities within academic libraries influence students' behavioral intentions to use libraries for academic purposes?
2. How does emotional solidarity with the library community affect students' behavioral intentions to engage in academic activities?
3. To what extent does social identity moderate the relationship between leisure activities, emotional solidarity, and students' behavioural intentions?
4. How does gender moderate the relationship between emotional solidarity and students' behavioural intentions to use the library for academic engagement?

To answer these questions, this study investigated the behavioural intentions of university students in Ghanaian academic libraries through the lens of leisure participation, emotional attachment, social identity, and gender. This inquiry advances our understanding of the psychosocial and contextual factors that affect academic engagement in library settings and provides actionable insights for improving library services to meet the needs of diverse student populations.

The remainder of this paper is organised as follows. Section 2 presents related literature, hypothesis development, and the theoretical framework of the study. Section 3 describes the materials and methods used in this study. The results of this study are presented in section 4. Section 5 presents a discussion of the key findings, and section 6 presents the conclusions and implications for policy actions.

2. Theoretical framework, literature review and hypotheses development

2.1. Theoretical framework

This study investigated the behavioural intentions of students in using academic libraries as predicted by their emotional solidarity, leisure activities, social identity, and gender using social identity theory, uses and gratifications theory, theory of planned behaviour, and the concept of gender differences.

2.1.1. Social identity theory

Social Identity Theory explains how people define themselves based on the groups they belong to, such as ethnic, professional, or academic communities (Kreuter & Lezin, 2002). In this study, the library was viewed as a social group in which students feel a sense of belonging. When students identify with a library community, they are more likely to participate in both academic and social activities. This sense of belonging increases emotional solidarity, which can strengthen students' intentions to use the library for academic purposes. Participation in group activities helps students feel more connected and valued, leading to stronger academic motivation (Fong et al., 2020; Mishra, 2020).

2.1.2. Uses and gratifications theory

The uses and Gratifications Theory explains why people choose certain environments or tools to meet their personal and social needs. In a library setting, students may use leisure activities to relax, connect with others, or enjoy informal learning. These experiences can improve their satisfaction with the library and encourage them to return for academic purposes (Williams, 2021). The more time students enjoy in the library, the more likely they are to view it as a valuable space for learning and personal development. The positive emotional experiences gained through leisure may support continued academic engagement.

2.1.3. Theory of Planned Behaviour

The Theory of Planned Behaviour focuses on how attitudes, social norms, and perceived control influence a person's actions (Ajzen, 2011). When students believe that using the library is supported by their peers and academic community, they may feel encouraged to follow this behaviour. A supportive environment can also effectively improve students' confidence in using the library. This increases the likelihood that they will form strong intentions to use the space for academic tasks (Mashroofa et al., 2023). The social and emotional environments of the library play a key role in this decision-making process.

2.1.4. Gender differences

Gender can influence how students respond to emotional and social

cues in a learning environment. Some studies suggest that female students are affected more by peer pressure and social norms in academic settings, including libraries (Zhang et al., 2023). When libraries create warm and inclusive environments, female students may feel more emotionally supported and motivated to use their academic resources. Gender differences may also influence students' experience of emotional solidarity. Female students who feel part of the library community may be more likely to engage in academic behaviours because of their responsiveness to emotional and social feedback.

2.2. Literature review and hypotheses development

2.2.1. Leisure in libraries and behavioural intentions to library

University libraries have changed in the recent years. However, they are no longer used for academic purposes. Many libraries now serve as social and leisure spaces that support both learning and student well-being. Recent studies have shown that students value environments that allow them to study, relax, and connect with others in one place (Okunlaya et al., 2022; Vedantham, 2020). This is especially true for Generation Z students, who are often drawn to spaces that offer enjoyable and emotionally engaging experiences (Janšto et al., 2024). Some studies have found that when students participate in leisure activities in libraries, such as attending events or relaxing in comfortable spaces, they are more likely to return to using the library for academic purposes (Barniskis, 2023; VanScoy et al., 2020). These positive experiences build emotional connections with the library and increase the motivation to study. For example, Mushtaq (Mushtaq et al., 2021) et al. (2021) observed that students who feel comfortable and satisfied in the library environment often use it more for academic activities.

Other researchers have supported these findings. Vedantham (2020) that when libraries offer flexible spaces for both leisure and learning, students feel more engaged and are more likely to use academic services. Okunlaya et al. (2022) also explained that comfortable library designs improve student motivation, especially when students are under academic stress. These studies suggest that leisure features in libraries can help to increase students' academic involvement. Outside the library setting, Janšto et al. (2024) studied the behaviour of Generation Z consumers when buying beverages. Their research showed that young people are influenced by emotional factors such as how a product looks or how it makes them feel. These ideas can be applied in libraries. When library spaces look welcoming and make students feel connected, they may be more likely to return and use the library for academic work. Therefore, we propose the following hypothesis.

H1. Higher engagement in leisure activities in the library leads to a stronger behavioural intention to use the library for academic purposes.

2.2.2. Emotional solidarity and behavioural intentions to library

The results indicated a significant association between students' heightened involvement with library resources and their deep feelings of affiliation with the library community. Upon visiting these premises, individuals not only have access to knowledge but also find a welcoming environment that promotes inclusivity through the library's friendly staff, possibilities for social contact, and hospitable ambience (Bladek, 2021). Students who feel this emotional solidarity not only have more positive evaluations of the library but also show a higher level of eagerness to use its academic resources. However, this association does not answer the crucial question of which specific factors intertwine segments of emotional solidarity (Kenyon & Henrich, 2024). Does a deep feeling of belonging to a community drive academic engagement or is the first impression of warmly welcomed the determining factor? Perhaps the motivation to delve further into the library's academic riches arises from the feeling of unity that one encounters with its fundamental principles (Kenyon & Henrich, 2024). Therefore, we propose the following hypothesis.

H2. Greater feelings of emotional solidarity within the library community lead to stronger behavioural intentions to use the library for academic purposes.

2.2.3. Moderation role of social identity

Students who form strong connections with the library community through active participation in events and organizations become part of their social identity (Rathbone et al., 2023). This sense of belonging leads to increased participation in leisure activities and strengthens emotional solidarity (Kišjuhas, 2023). The library environment fosters emotional solidarity and motivates students to participate in academic activities (DeVille & Sughrue, 2023). The welcoming environment and helpful personnel also contribute to this sense of belonging. The concept of social identity posits that individuals derive a significant portion of their self-concept from their associations with particular social organizations (Froehlich et al., 2023). The attitudes, behaviours, and motivations of individuals are profoundly impacted by group identification, which in turn affects their interactions both within the group and in external contexts (Zhu et al., 2021). In the domain of libraries, the academic dedication of students who exhibit a strong sense of community membership through participation in study groups, organizations, or events may be more significantly impacted by leisure pursuits and emotional solidarity (Wojciechowska, 2023). Participating in leisure activities at the library is not merely a means of occupying time for several students; rather, it reinforces their commitment to the community's shared values and objectives, and fosters a stronger sense of community. Based on the literature review, the following hypotheses were formulated.

H3a. The moderating effect of social identity influences the relationship between leisure and behavioural intentions.

H3b. The moderating effect of social identity influences the relationship between emotional solidarity and behavioural intentions.

2.2.4. Moderating role of gender

Dynamic correlation between gender and the impact of emotional unity on academic ambition. While there is significant evidence indicating that females are more inclined than males to use library resources such as leisure spaces (Fong et al., 2020), an important question remains: Does this gender imbalance in library community involvement have the same effect on students' academic motivation as emotional connections within the library community? Studies suggest that when it comes to library utilisation, females may be more susceptible to social influences and cultural expectations (Cox, 2021). This raises the interesting concept that students' tendency to use the library for academic reasons may be strongly affected by the welcoming atmosphere, helpful staff, and emotional bonds they feel with other library users (Zhou, 2022). A strong sense of community within a library may successfully encourage and motivate female clients to make better use of their academic resources. Thus, this study examines the following hypotheses.

H4. The positive relationship between emotional solidarity and behavioural intentions is moderated by gender

3. Materials and methods

3.1. Research design

We employed a quantitative research approach by adopting a cross-sectional research design to examine the effects of leisure activities and emotional solidarity on behavioural intentions towards the use of academic libraries among Ghanaian undergraduate and graduate students. The quantitative design makes it possible to establish statistically significant relationships between variables and draw reliable generalizations, thereby offering valuable insights into the dynamics of library usage in the Ghanaian context (Samii, 2016). The current study encompasses a broad range of participants, including undergraduate and

postgraduate students enrolled in 20 public universities in Ghana, including traditional public universities and technical universities. A representative sample was obtained using stratified sampling to reflect the distribution of the student body accurately. The 20 universities were stratified into distinct subgroups (strata) based on pertinent characteristics, specifically the distinction between traditional public universities and technical universities. The sampling technique considers the size of each stratum and samples respondents proportionally. To augment the representativeness of the sample, the participants were asked to recommend colleagues from the same university until the desired sample size was achieved. We were able to centrally locate and recruit volunteers within any academic institution. An average of 50 students were selected from each of the 20 institutions for 1000 participants in the initial target sample. Following the data collection, 838 students comprised the final sample used for the analysis. This corresponds to an approximate response rate of 84 %, which is deemed acceptable (Louangrath, 2014). The study employed various statistical tools, including Microsoft Excel for data cleaning and preliminary analysis and SPSS version 26 and Stata version 16 for descriptive statistics, hypothesis testing, and advanced regression analysis.

3.2. Data collection and analysis

The study data were collected over the entire academic semester, spanning from August 17, 2023, to January 2, 2024, after ethical clearance was obtained (ethical clearance number HTU/DRI/EC2023-026). This technique was selected to provide a uniform representation of student library use across different timeframes. To guarantee accessibility and inclusiveness among target demographics, researchers have used a comprehensive approach that recognizes many technological contexts. Google Forms surveys were systematically delivered to the students' bodies through WhatsApp and email. This digital platform facilitated efficient connections with students, including diverse geographical locations and degrees of Internet access across all 16 regions of Ghana. The demographic information of the respondents is presented in Table 1.

The study was conducted across public academic institutions in Ghana, with participants drawn from diverse academic disciplines and all 16 regions, ensuring broad representation. A total of [insert final sample size] students participated, with the majority (over 66 %) aged between 20 and 29 years, reflecting the library's critical role in early academic life. Female respondents comprised 63.7 % of the sample, while males accounted for 36.3 %, indicating a need to consider gender-based preferences in library engagement. Most participants were undergraduates (49 %), followed by professional training programs (30.7 %) and a smaller proportion of postgraduate students. This demographic distribution aligns with the study's focus on leisure activities, emotional connections, social identity, and gender, and provides a comprehensive understanding of library usage patterns within the Ghanaian higher education context.

Table 2 indicates that library visits were infrequent for most students, emphasizing the need for flexible resources. The primary reasons for the visits were academic pursuits and borrowing materials. While many find the library conducive to study, some experience discomfort. Although not widely perceived as a leisure space, a notable group views it as such, suggesting opportunities to support both academic and leisure activities. These findings highlight the importance of accessible materials and tailored environments to enhance student engagement.

Table 1
Participants demographics and settings.

		Frequency	%
Gender	Female	534	63.7
	Male	304	36.3
Age	Below 20	54	6.4
	20–29 years	553	66.0
	30–39 years	186	22.2
	40–49 years	39	4.7
	50–59 years	6	0.7
Academic level	Certificate	8	1.0
	Diploma	257	30.7
	Higher National Diploma (HND)	66	7.9
	Undergraduate	411	49.0
	Posgratduate	52	6.2
	Professional Degree	44	5.2
Major	Business Programs	122	14.6
	Health Programs	108	12.9
	STEM Programs	291	34.7
	Arts and Humanities Programs	317	37.8
N = 838			

Table 2
Psychographics of participants.

		Frequency	%
Frequency of visits to academic library	Never	43	5.1
	Rarely	229	27.3
	Occasionally	411	49.0
	Frequently	139	16.6
	Very frequently	16	1.9
	Mean = 2.83; S.D = 0.832		
Purpose of visit to the academic library	Study and research	592	70.6
	Borrowing books or materials	133	15.9
	Attending workshops or events	27	3.2
	Using computer or printing facilities	86	10.3
Satisfaction with the academic library's facilities	Very dissatisfied	37	4.4
	Dissatisfied	85	10.1
	Neutral	327	39.0
	Satisfied	317	37.8
	Very satisfied	72	8.6
	Mean = 3.35; S.D = 0.965		
Satisfaction with the academic library's services	Very dissatisfied	34	4.1
	Dissatisfied	56	6.7
	Neutral	330	39.4
	Satisfied	357	42.6
	Very satisfied	61	7.3
	Mean = 3.36; S.D = 0.933		
Feeling comfortable studying in the academic library?	Not at all comfortable	92	11.0
	Somewhat comfortable	33	3.9
	Moderately comfortable	302	36.0
	Very comfortable	110	13.1
	Extremely comfortable	301	35.9
	Mean = 3.59; S.D = 0.874		
Consideration of academic library as a leisure activity	Strongly disagree	51	6.1
	Disagree	122	14.6
	Neutral	286	34.1
	Agree	329	39.3
	Strongly agree	50	6.0
	Mean = 3.24; S.D = 0.981		
N = 838			

3.3. Measurement of variables

Behavioural intention: We assessed students' potential involvement with the library in the future by gauging their probability of using the library for different academic objectives. We used statements such as " I am likely to borrow books or articles from the library for my upcoming research project" using a rating system ranging from 1 to 5 on a Likert scale. Based on research by (Davis, 1989; DeVille & Sughrue, 2023), these findings provide a clear understanding of how students plan to use the library and highlight the influence of several elements on their actual behaviour.

Leisure activities: The library's holdings extend beyond the printed materials. We explored how leisure activities influence the development of students' library experience. Statements such as " I often visit or use the university library for leisure with friends or classmates." The measurement followed Bladek (2021), who examined the effects of incidental library usage on the overall perception and level of engagement of individuals. Therefore, it is possible to identify strategies to foster a more welcoming and livelier environment for each student.

Emotional solidarity: A strong feeling of belonging to the library community may greatly incentivise users to engage. This concept is measured by statements such as " I feel a sense of attachment or belonging to the university library." Using knowledge gained from prior research, such as (Kenyon & Henrich, 2024), we used survey tools to evaluate the emotional connections that students establish with the library.

Social identity: To further explore the extent to which students feel connected, we modified the well-known social identity questionnaire (GholamiNezhad et al., 2022) to suit the library's purpose, with Likert scale items (1–5): 'I believe that engaging in leisure activities at the academic library enhances my creativity and self-expression?'

Table 3 shows that individuals exhibited a moderately positive inclination towards using the library for academic purposes, with an average score of 3.79 for behavioral intentions. Leisure activities at the library received a moderate average score of 3.56, while emotional solidarity with the library community was slightly above neutral at 3.73, with a mean score of 3.66, indicating a moderate emotional connection. Gender distribution among participants showed that 36.3 % were male and 63.7 % were female, suggesting a potential moderating effect of gender on the relationship between emotional solidarity and behavioural intentions (see Table 4).

The findings show that positive experiences, belonging, and shared identity enhance students' motivation to use academic libraries, fostering both a community and academic focus. Leisure activities strengthen social cohesion and indirectly promote academic engagement. Males visit libraries more often than females, and higher education levels correlate with increased library use (Zhang et al., 2023). Engagement varies according to academic major, highlighting the need for tailored strategies. The strong link between positive library experiences and behavioural intentions underscores the importance of improving services to create a supportive and welcoming academic environment.

Table 3
Variable definition, measurement and descriptive statistics.

Variable	Notation	Definition	Unit	Mean	S.D.	Obs.
Behavioural intention	BehI	Factorial measure of behavioural intentions	Quantity	3.790	0.714	838
Leisure activities	Lei	Factorial measure of leisure activities	Quantity	3.559	0.737	838
Emotional solidarity	EmS	Factorial measure of emotional solidarity	Quantity	3.727	0.724	838
Social identity	SocI	Factorial measure of social identity	Quantity	3.662	0.705	838
Gender	Gender	Gender of students (1 = male; 0 = female)	Quantity	0.363	0.481	838

3.4. Factor analysis

Table 5 presents the results of the factor analysis, identifying four key constructs influencing students' behavioural intentions to use academic libraries: behavioural intention, leisure activities, emotional solidarity, and social identity. Behavioural intention accounted for 68.929 % of the variance, with factor loadings ranging from 0.676 to 0.880. Leisure activities explained 52.610 % of the variance with loadings between 0.523 and 0.816. Emotional solidarity explained 68.896 % of the variance, with strong loadings between 0.700 and 0.891. Social identity accounted for 58.901 % of the variance with loadings ranging from 0.504 to 0. 829.The reliability and validity were confirmed using multiple tests. The KMO values for all constructs were above 0.80, indicating adequate sampling. Bartlett's test of sphericity was significant ($p < 0.001$), confirming suitability for factor analysis. The Cronbach's alpha values ranged from 0.812 to 0.907, demonstrating high internal consistency. These results confirm the robustness and reliability of the measurement model.

3.5. Model specification

The study used multiple regression and moderation analyses by employing an ordered probit regression model to analyse the relationship between leisure activities (LeiA), emotional solidarity (EmS), social identity (SocI), and gender and students' behavioural intentions (BehI) to use the library for academic purposes. The model utilises discrete ordinal values to represent (BehI), which consists of the following categories: "very likely", to "unlikely", unlikely. Emotional solidarity and leisure activities are regarded as independent variables represented by $X_1, X_2 \ldots X_n$, whereas social identity and gender serve as moderating variables and academic level (AcL), major (Mj), and satisfaction with library facilities and services (SLFS) are the control variables. The model posits the presence of fundamental threshold values (cutoff points) that differentiate between different forms of behavioural intentions (BehI). The thresholds are represented by $T_1, T_2, \ldots, T_{(K-1)}$, which K constitute the total number of groupings. The thresholds are affected by an unobservable latent variable, represented as, Y^* which is also affected by the latent variable. The association between an observed dependent variable Y and a latent variable Y^* is explained by threshold values and error terms. Thus, $1, 2, \ldots, (K-1)$ is equated to signify the error terms linked with the threshold.

The empirical models for this study were estimated as follows. First, we estimate the direct-ordered logistic regression model, followed by the moderation-ordered logistic regression model.

Main effect model

$$Logit(P(BehI > k \mid X)) = \beta_0 + \beta_1 Lei + \beta_2 EmS + \beta_3 AcL + \beta_4 Mj + \beta_5 SLFS + \varepsilon \quad (1)$$

Moderation effects model

$$Logit(P(BehI > k \mid X)) = \beta_0 + \beta_1 Lei + \beta_2 EmS + \beta_3 AcL + \beta_4 Mj + \beta_5 SLFS + \beta_6 Lei{*}SocI + \beta_7 EmS{*}SocI + \beta_7 EmS{*}Gender + \varepsilon \quad (2)$$

Where.

Table 4
Correlation matrix.

	BehI	Lei	EmS	SocI	Gender	AcL	Mj	SLFS
BehI	1							
Lei	0.601**	1						
EmS	0.674**	0.464**	1					
SocI	0.718**	0.647**	0.613*	1				
Gender	0.618*	0.091	0.150*	0.082	1			
AcL	0.412**	0.004	0.068	0.008	0.322*	1		
Mj	0.724*	0.026	0.096	0.041	0.240*	0.111*	1	
SLFS	0.837**	0.248	0.336*	0.226*	0.113*	0.044	0.07	1

Note(s): **p < 0.05, *p < 0.1. BehI: behavioural intentions, Lei: leisure activities, EmS: emotional solidarity, SocI: social identity, AcL: academic level, Mj: major, SLFS: satisfaction with library facilities and services.

Table 5
Results of factors analysis.

Factor name	Variables	Factor loading	% of Variance explained	Cronbach's alpha
Behavioural intention	BehI1	0.814	68.929	0.906
	BehI2	0.880		
	BehI3	0.872		
	BehI4	0.855		
	BehI5	0.865		
	BehI6	0.676		
	KMO = 0.880			
	Bartlett's test of sphericity	Chi-squared = 3404.786	p-value = 0.000	
Leisure activities	Lei1	0.714	52.610	0.812
	Lei2	0.816		
	Lei3	0.707		
	Lei4	0.762		
	Lei5	0.792		
	Lei6	0.523		
	KMO = 0.824			
	Bartlett's test of sphericity	Chi-squared = 1675.969	p-value = 0.000	
Emotional solidarity	EmS	0.835	68.896	0.907
	EmS	0.820		
	EmS	0.874		
	EmS	0.891		
	EmS	0.846		
	EmS	0.700		
	KMO = 0.908			
	Bartlett's test of sphericity	Chi-squared = 3169.637	p-value = 0.000	
Social identity	SocI1	0.829	58.901	0.857
	SocI2	0.822		
	SocI3	0.784		
	SocI4	0.672		
	SocI5	0.775		
	SocI6	0.504		
	KMO = 0.847			
	Bartlett's test of sphericity	Chi-squared = 2151.156	p-value = 0.000	

Y (BehI): Dependent variable, behavioural intentions to use the library for academic purposes (ordinal).
X: Vector of independent and control variables.
β: Estimated coefficients.
ε: Error term.
k represents the kth category of ordered dependent variable.

Interaction terms capture moderating effects (e.g., Lei * SocI, EmS * SocI, EmS * Gender).

4. Results

The results of the ordered logistic and ordered probit regressions, shown in Tables 6 and 7, indicate a good model fit and highlight the significant influence of the study variables on behavioural intentions. Table 6 presents the results without the interaction effects of social identity and gender, while Table 7 includes these interaction effects (see Table 8).

Table 6
Results of ordered logistics and probit regressions on leisure activities, emotional solidarity and behavioural intention.

Dependent variable: Behavioural intentions				
Variable	Ordered Logistics		Ordered probit	
	Lei	EmS	Lei	EmS
Lei	1.271***		0.990***	
	(11.69)		(17.96)	
EmS		2.160***		1.303***
		(17.06)		(21.31)
AcL	0.199**	0.219***	0.109**	0.376***
	(2.84)	(3.82)	(3.00)	(9.72)
Mj	−0.134**	−0.128**	−0.253***	−0.056*
	(-2.25)	(-2.17)	(-6.66)	(-1.65)
SLFS	0.514***	0.667***	0.119**	0.378***
	(7.59)	(9.36)	(3.04)	(9.79)
Cut1(*strongly disagree*)	1.193	2.454	0.784	1.249
Cut2(*disagree*)	1.328	2.630	0.851	1.325
Cut4(*agree*)	2.577	4.113	1.509	1.998
Cut5(*strongly agree*)	2.622	4.164	1.534	2.022
Psedo R^2	0.639	0.532	0.346	0.442
Log-Likelihood	−1842.574	−1913.898	−2010.679	−1939.009
LR χ^2 Test	711.03***	568.38***	374.82***	518.16
AIC	3735.148	2137.867	4069.359	4423.272
BIC	3853.423	2156.791	4182.903	4536.817
Obs.	838	838	838	838

Note (s):***p < 0.001, **p < 0.05, *p < 0.1. The Z-statistics are shown in parentheses. BehI: behavioural intentions, Lei: leisure activities, EmS: emotional solidarity, SocI: social identity, AcL: academic level, Mj: major, SLFS: satisfaction with library facilities and services.

4.1. Results of baseline ordered logistics regression without moderating variables

The findings in Table 6 indicate that students' behavioural intentions to use the library were influenced by several factors. Leisure activities, such as attending seminars or joining study groups, increased the likelihood of library use for academic purposes 1.271-fold. Emotional solidarity within the library also plays a crucial role, with each incremental increase in students' emotional cohesion boosting their intention to use the library by a factor of 2.160. Furthermore, there was a positive relationship between students' academic level and their intention to use library resources, with higher academic demands leading to increased usage. Students' academic major also significantly affected their library usage intentions, reflecting varying expectations across disciplines. Satisfaction with library facilities and services was a key determinant,

Table 7
Results of ordered logistics and probit regressions on leisure activities, emotional solidarity, social identity, gender and behavioural intention.

Dependent variable: Behavioural intentions				
Variable	Ordered Logistics		Ordered probit	
	Lei	EmS	Lei	EmS
Lei	1.933		1.148***	
	(4.48)		(4.89)	
EmS		0.447***		0.247***
		(3.97)		(4.25)
SocI	0.733*	0.919***	2.319***	7.638***
	(1.75)	(29.04)	(5.23)	(30.33)
SocI × Lei	−0.356**		−0.210**	
	(-3.01)		(-3.25)	
SocI × EmS		0.592***		0.335***
		(13.81)		(14.18)
Gender		3.736***		2.160***
		(4.68)		(33.03)
Gender × EmS		−0.969		−0.587***
		(-4.68)		(-5.10)
AcL	0.965***	0.311***	0.257***	0.505***
	(4.66)	(3.67)	(12.98)	(32.54)
Mj	−0.123**	−0.112*	−0.068*	−0.111
	(-2.03)	(-1.79)	(-1.93)	(-2.71)
SLFS	0.600***	0.308***	0.401***	0.237***
	(13.83)	(3.62)	(6.18)	(5.31)
Cut1(*strongly disagree*)	4.598	0.816	2.409	0.242
Cut2(*disagree*)	4.804	0.989	2.529	0.667
Cut4(*agree*)	6.601	2.469	3.543	1.488
Cut5(*strongly agree*)	6.672	2.527	3.579	1.783
Psedo R^2	0.721	0.645	0.683	0.3253
Log-Likelihood	−1755.610	−1788.747	−1765.316	−1503.3027
LR χ^2 Test	884.96***	818.68***	865.55***	1449.68***
AIC	3568.265	3025.452	3588.631	3060.605
BIC	3691.271	3148.458	3725.831	3188.343
Obs.	838	838	838	838

Note (s):***p < 0.001, **p < 0.05, *p < 0.1. The Z-statistics are shown in parentheses. BehI: behavioural intentions, Lei: leisure activities, EmS: emotional solidarity, SocI: social identity, AcL: academic level, Mj: major, SLFS: satisfaction with library facilities and services.

Table 8
Hypothesis testing results.

Hypothesis	Statement	Supported?	Statistical Test	p-value
H1	Leisure activities → Behavioural intention	Yes	Ordered Logistic Regression	<0.01
H2	Emotional solidarity → Behavioural intention	Yes	Ordered Logistic Regression	<0.01
H3a & H3b	Social identity moderates the relationship between leisure/emotional solidarity and behavioural intention	Yes	Ordered Logistic Regression with Interaction Terms	<0.01
H4	Gender moderates the effect of emotional solidarity on behavioural intention	No	Ordered Logistic Regression with Interaction Terms	>0.05

with a 0.514 increase in satisfaction, significantly enhancing the likelihood of students intending to use the library more frequently. These findings emphasise the importance of creating an inclusive, well-equipped library environment to cater to diverse student needs.

4.2. Results of ordered logistic regression with moderating variables (social capital and gender)

Table 7 analyses the connections between leisure activities, emotional solidarity, and students' intentions to use the library, considering the moderating roles of social capital and gender. The findings indicate that higher social capital within the library community correlates positively with students' likelihood of using the library for academic purposes, emphasizing the need for an inclusive environment that fosters social interactions. However, the positive impact of leisure activities on library usage intentions was slightly reduced for those with high social capital, suggesting that strong social connections may lead individuals to prioritise academic tasks over leisure. The interaction between emotional solidarity and social capital creates strong synergy, with individuals who possess both demonstrating the highest intention to use the library academically. Additionally, males showed greater behavioural intentions towards library usage compared to females, highlighting potential gender-specific barriers. Particularly, for women with strong emotional ties to the library, the gender gap in usage intentions narrows, indicating the importance of emotional solidarity in addressing these disparities.

4.3. Sensitivity analysis and robustness testing using ordered probit regression

The ordered probit regression findings align closely with those of the baseline-ordered logistic regression, indicating consistent relationships between significant factors and behavioural intentions. The key determinants of library usage for academic purposes include student satisfaction with library facilities, academic standing, leisure activities, and emotional cohesion. Prioritising these factors can significantly enhance the library usage. Sensitivity and robustness tests highlighted the need for a holistic approach to boost students' motivation for library use. Libraries should focus on promoting emotional solidarity, facilitating leisure activities, enhancing social capital, and addressing gender inequalities to create dynamic learning environments that support diverse student needs, ultimately aiding academic achievement and personal growth. This study underscores the importance of fostering inclusivity and diversity in library utilisation among students.

5. Discussion

Our findings emphasise the significance of leisure activities in bringing about significant changes. The incorporation of study groups, seminars, and game evenings into the library environment provides recreational opportunities and fosters a sense of community and belonging. This expands the library's role beyond academic pursuits and aligns with the evolving concept of libraries as dynamic centres for social interaction, as highlighted in (Bladek, 2021). Libraries can attract students who are not solely interested in academic resources, but also seek opportunities for social engagement and recreational activities. Emotional solidarity is an additional critical element. The findings of our research, in conjunction with the investigations conducted by (Kenyon & Henrich, 2024) and (DeVille & Sughrue, 2023), underscore the importance of establishing an inviting and inclusive library environment to promote academic engagement. Developing an atmosphere of inclusivity and establishing an emotional bond creates a favourable setting in which students are motivated to utilise resources at their disposal. Promoting inter-student engagement and implementing strategies such as peer mentoring programs can significantly contribute to the development of this vital sense of unity.

Our study establishes a clear correlation between educational attainment and inclination to utilise library resources. This finding is consistent with previous research (Mushtaq et al., 2021), which also suggests that students with greater academic obligations are more likely to rely on library resources. There are several possible explanations for

this phenomenon, including the increased reliance of students on the library for academic support, recognition of the library's role in their success, and their expanding knowledge of library resources as they progress in their academic pursuits. The research also revealed notable inconsistencies in the intention to use libraries among different academic subjects, indicating potential variations in expectations and prerequisites across disciplines. These findings align with research conducted by (Kišjuhas, 2023), which emphasizes the influence of academic communities and social identity on library usage. Libraries can tailor their services and resources to foster stronger partnerships with student organizations and cater to the unique needs of different academic fields. Thus, our research validates the critical importance of student satisfaction with resources and services provided by libraries. As highlighted by (Bladek, 2021) and (Wojciechowska, 2023), libraries must provide an environment that is well maintained, fully functional, and adequately furnished. The findings of this study indicate a significant positive correlation between heightened satisfaction and a greater likelihood of students utilising libraries more frequently. This underscores the significance of libraries for high-quality provisions.

Our study not only ascertains the direct influence of leisure activities and emotional solidarity on the behavioural intention to use libraries but also investigates the interaction of social capital and gender on these relationships. A significant finding is the importance of social identity in the context of the library community. The intention of students to use the library was positively correlated with increased levels of social identity, which was fostered through the creation of an inclusive and welcoming environment. This aligns with the results reported by (Kišjuhas, 2023) and (Rathbone et al., 2023), which emphasise the influence of academic community membership and social identity on library utilisation. To take advantage of this opportunity, libraries may promote social interaction through various means, including the organisation of events, facilitation of study groups, and provision of pleasant environments. The purpose of these initiatives is to foster a sense of camaraderie and support among students in library environments.

Our findings also revealed that social identity moderated the relationship between library usage and leisure activities. Participation in extracurricular pursuits frequently increases students' propensity to utilise the library; however, this effect is marginally diminished for individuals who possess considerable social capital. Strong social ties, according to (Wojciechowska, 2023), may suffice to fulfil a portion of one's needs through social engagement. Consequently, academic pursuits may be given precedence over leisure library excursion. This finding highlights the importance of tailoring leisure activities to accommodate individuals with varying levels of social capital. For instance, students with greater social connections may find it more conducive to engage in concentrated studies at the library, which could be accomplished by offering intellectually stimulating leisure activities.

Synergy is produced when emotional solidarity and social capital exert combined influence. Individuals who exhibit the dual nature of robust social networks and a deep emotional affinity towards the library demonstrate the greatest degree of motivation to utilise it for scholarly endeavours. Additionally (DeVille & Sughrue, 2023), emphasise the importance of an inclusive and supportive environment in promoting scholarly involvement. Libraries can foster an exceptionally conducive environment for academic success. This may encompass initiatives, such as collaborative study spaces, peer mentoring programs, and activities that cater to a diverse array of student interests, thereby fostering academic motivation and social cohesion.

Essentially, our findings raise intellectually stimulating questions regarding gender-based disparities in library utilisation. Males exhibit a significantly higher propensity to utilise libraries than females, and this disparity persists even when other variables are considered. This highlights the importance of adopting targeted strategies to address potential challenges specific to female students, including the lack of safe spaces, discomfort in study areas dominated by males, and cultural conventions regarding library usage. A significant correlation was found between gender and affective unity. Gender differences in the behavioural intentions of women with strong emotional attachment to the library were significantly diminished. This suggests that fostering an environment of inclusivity and forming emotional connections within the library could potentially serve as an effective approach to promote gender equality in library utilisation. Libraries may implement strategies such as designating study areas exclusively for female patrons, providing mentoring programs for female students, and arranging academic achievement-oriented sessions that accommodate the needs of women. By gaining an understanding of these complex interrelationships and mitigating factors, libraries can move beyond a standardised approach and implement targeted strategies to cater to the diverse needs and preferences of patrons. It is of utmost importance to create an environment that fosters social unity, emotional assistance, and gender equality to maximise library usage and ensure that all students have equal access to academic resources and support.

5.1. Research significance and implications

Our study highlights the importance of library administration and future research to understand the factors that influence students' intentions to use libraries. The findings emphasise the significance of having easily accessible scholarly resources, but also delve into the social and affective aspects of the library experience. To encourage students to use libraries, it is crucial to create an environment that promotes community engagement, active engagement, and a sense of belonging. These insights have implications for both library administrators and researchers. The incorporation of recreational activities into libraries extends beyond providing leisure opportunities. It also creates a sense of community and belonging, transforming the library into a space that extends beyond its academic purposes. This aligns with (Bladek, 2021) perspective that libraries are becoming vibrant hubs of social engagement. Research conducted by (Kenyon & Henrich, 2024) further emphasizes the importance of creating an inviting and inclusive environment to encourage academic participation. Libraries have the potential to attract students who seek not only academic resources, but also opportunities for social interaction and recreational pursuits, thus enhancing their overall learning experience.

This study highlighted the importance of effective cohesion in community libraries. These findings support previous research conducted by (DeVille & Sughrue, 2023), which also emphasizes the significance of creating an inclusive and welcoming environment in libraries to promote scholarly engagement. Fostering inclusivity and establishing emotional connections will make students more likely to utilise the available resources. Implementing strategies, such as peer mentoring programs and encouraging interactions among students, can greatly enhance the development of a strong sense of solidarity, leading to increased motivation and academic success. Social capital has emerged as a crucial factor, as highlighted by (Rathbone et al., 2023). This study revealed a positive correlation between higher levels of social capital within a library community and the likelihood of students using libraries for academic purposes. This underscores the importance of cultivating an inclusive and welcoming environment that promotes social interaction and a sense of belonging. The findings also suggest that social capital moderates the relationship between library use and leisure activity. Therefore, it is necessary to provide tailored recreational pursuits that accommodate individuals with varying degrees of social integration.

6. Conclusion

6.1. Research conclusions

In conclusion, our study highlighted the importance of considering gender, social capital, affective solidarity, and leisure activities when determining students' library utilisation. Incorporating recreational

activities, creating an inclusive environment, and adapting services will help to meet diverse social needs. Therefore, libraries can foster a sense of community and academic engagement among their students. These findings have significant implications for library design and administration as well as for future research, with the ultimate aim of improving libraries as active and equitable spaces that positively impact student academic performance.

6.2. Limitations and future research directions

This study is limited by its cross-sectional design, reliance on self-reported data, and focus on Ghana, which may affect generalisability. Future research should use longitudinal designs, explore cross-cultural contexts, and examine how library design, technology, and staff support influence student behaviour, including gender-based and social interaction differences.

CRediT authorship contribution statement

Vincent Kweku Asimah: Writing – review & editing, Writing – original draft, Supervision, Methodology, Data curation, Conceptualization. **Amy Pascalia Abra Asimah:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Data curation.

Statement of informed consent

Before administering the questionnaire, consent was obtained from the participants, and ethical clearance was secured from the Ho Technical University Research Ethics Committee on May 12, 2023 (ethical clearance number HTU/DRI/EC2023-026). Furthermore, the participants were duly notified that the data they provided would be kept private and confidential.

Data availability statement

The data will be made available upon request from the corresponding author.

Declaration of competing interest

The authors declare that they have no affiliations with or involvement in any organisation or entity with any financial interest in the subject matter or materials discussed in this manuscript.

Ethics statement

This study titled ***"Dynamics of Behavioural Intentions in Academic Libraries: The Effect of Leisure, Emotional Solidarity, and the Moderating Role of Social Identity"*** was conducted in strict adherence to ethical research standards to ensure the integrity of findings and the respectful treatment of participants. Ethical considerations were integrated at each stage of the research process.

Prior to data collection, ethical approval was obtained from Ho Technical university, Directorate of Research and Innovation with reference number HTU/DRI/EC2023-026. Participation in the study was voluntary, and all participants were informed about the study's objectives, procedures, and potential impacts. Participants provided informed consent, acknowledging their right to withdraw at any stage without penalty and understanding that participation entailed no foreseeable risks to their wellbeing.

To maintain confidentiality, no personally identifiable information was collected or stored, and all data were anonymized before analysis. The responses were securely stored in compliance with data protection regulations, and only the research team had access to this information.

This research was conducted in alignment with the principles of the Declaration of Helsinki, respecting participants' rights, autonomy, and privacy. The findings of this study are reported transparently and objectively, with no conflicts of interest or undue influence affecting the outcomes.

By ensuring these ethical protocols, the study aims to contribute to an authentic understanding of behavioural intentions in academic libraries, examining the intersections of leisure, emotional solidarity, and social identity without compromising ethical integrity.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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